

MoReAD: Cross-Representation Mixture-of-Experts and Conditional Recovery for Unsupervised Industrial Anomaly Detection

Anonymous submission

Abstract

Unsupervised industrial anomaly detection requires accurate modeling of normal data using only defect-free training samples. However, in complex industrial scenarios, incomplete perceptual representations and ambiguous hierarchical recovery cause reconstruction errors to reflect both legitimate normal variations and true defects, thereby reducing localization reliability. In this work we propose a unified feature-reconstruction framework to address these issues. First, the Cross-Representation Perceptual Coherence Mixture-of-Experts (CR-MoE) retains DINO as the primary reconstruction and anomaly-measurement space, while using EUPE to evaluate low-dimensional expert candidates during training. The resulting cross-representation selection knowledge is distilled into a lightweight router. Second, the Conditional Basis Transition Module (CBTM) generates constrained transition bases from the current reconstructed state at each decoding stage and predicts low-dimensional combination coefficients from the corresponding shallow features. This design models sample-specific normal recovery paths while avoiding information shortcuts caused by direct feature transmission. At inference, EUPE and all auxiliary evaluation branches are removed. Compared with the strongest baseline on each dataset, it improves P-AUPRO@0.05 by 2.4 percentage points on average, corresponding to a relative gain of 3.2%, and establishes new state-of-the-art localization across all five benchmarks.

Introduction

Industrial anomaly detection (IAD) learns from normal data and detects unseen defects through feature-reconstruction discrepancies (Liu et al. 2024; Bergmann et al. 2019; Zou et al. 2022; Deng and Li 2022; Guo et al. 2025). Production variation in illumination, texture, text, layout, and manufacturing tolerances (Wang et al. 2024; Heckler-Kram et al. 2026) can make normal content appear anomalous. The central problem is to **account for legitimate normal variation while preserving poor reconstruction of anomalies.**; we target representation incompleteness under a fixed metric and recovery ambiguity after compression.

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Representation incompleteness under a fixed anomaly metric arises because DINO provides structural awareness and dense correspondence (Caron et al. 2021; Oquab et al. 2024; Guo et al. 2025), but may omit text, layout, boundaries, and task-dependent cues. EUPE offers broader multi-teacher knowledge (Zhu et al. 2026), yet replacement or fusion alters DINO’s scoring geometry. Our **Cross-Representation Perceptual Coherence Mixture-of-Experts (CR-MoE)** keeps DINO as the sole reconstruction and anomaly space and EUPE as training-only supervision. Low-capacity experts propose DINO-space corrections; joint DINO–EUPE evaluation yields responsibilities distilled into a DINO-only router. EUPE branches are removed at inference.

Hierarchical recovery ambiguity arises because compressed deep features do not uniquely determine shallow states: deterministic decoders average valid paths, while unrestricted encoder skips may transmit anomalies (Bishop 1994; Sohn, Lee, and Yan 2015; Yong and Brintrup 2022). Our **Conditional Basis Transition Module (CBTM)** replaces direct transfer with compact, sample-conditioned transitions: the decoder state generates basis directions, and the corresponding encoder feature predicts bounded K_b -dimensional coefficients.

Together, CR-MoE and CBTM model normal variation through **condition-selected, capacity-constrained transformations**. Without anomalies, pseudo anomalies, or extra inference encoders, they improve low-false-positive localization across industrial benchmarks.

Our contributions are:

- We identify representation incompleteness and recovery ambiguity as coupled causes of unreliable feature reconstruction.
- CR-MoE transfers perceptual knowledge through cross-representation evaluation while retaining DINO-only inference.
- CBTM models layer-wise recovery through capacity-constrained conditional basis transitions.

Related Work

Industrial anomaly detection under real-world variation. IAD models defect-free data and detects deviations at test time (Liu et al. 2024; Tao et al. 2022; Xie et al. 2024).

MVTec AD and VisA approach saturation, whereas Real-IAD and MVTec AD 2 expose illumination shifts, reflective or transparent objects, fine textures, and broad normal variation (Bergmann et al. 2019; Zou et al. 2022; Wang et al. 2024; Heckler-Kram et al. 2026). Robust IAD must therefore suppress false alarms on legitimate structures without missing defects.

Reconstruction-based industrial anomaly detection. Normal-trained models are expected to recover anomalies poorly. Image methods use corruption or generative restoration (Zavrtanik, Kristan, and Skocaj 2021; Beizae et al. 2025); feature methods use bottlenecks, loose reconstruction, domain adaptation, or normal prototypes (Salehi et al. 2021; Deng and Li 2022; Tien et al. 2023; Guo et al. 2023, 2025; Liu et al. 2025; Luo et al. 2025). Training-free RAD instead retrieves context-conditioned multi-level normal features (Zhang et al. 2026). Reconstruction constraints preserve anomaly sensitivity but may discard legitimate details.

Representation bottlenecks and cross-representation transfer. Information-bottleneck theory captures the reconstruction tension: compression suppresses shortcuts, but downstream transformations cannot recover discarded factors (Tishby, Pereira, and Bialek 1999; Tishby and Zaslavsky 2015; Alemi et al. 2017). Self-supervised features likewise inherit invariances from their pretraining views (Tian et al. 2020). DINO and DINOv2 offer semantic structure and dense correspondence but can miss industrial details, whereas substituting or fusing broader multi-teacher EUPE features changes the anomaly-measurement geometry (Caron et al. 2021; Oquab et al. 2024; Amir et al. 2021; Zhu et al. 2026). Privileged distillation and mixture-of-experts support conditional transfer (Vapnik and Vashist 2009; Lopez-Paz et al. 2016; Jacobs et al. 1991; Shazeer et al. 2017), yet do not preserve a fixed anomaly space while using a heterogeneous encoder solely to evaluate capacity-constrained corrections, motivating CR-MoE.

Conditional hierarchical feature recovery. Compression also makes recovery one-to-many: deterministic decoders average valid shallow states, whereas unrestricted skips may transmit anomalies (Geirhos et al. 2020; Yong and Brintrup 2022). Conditional generation and mixture-density prediction address such mappings (Bishop 1994; Sohn, Lee, and Yan 2015), and basis regularization limits transition capacity (Bansal, Chen, and Wang 2018; Cogswell et al. 2016), but feature-reconstruction IAD still uses fixed transitions or direct feature injection. Representation incompleteness and recovery ambiguity therefore remain coupled under realistic variation, motivating CBTM with CR-MoE.

Method

As illustrated in Figure 1, our framework is built upon the loose feature-reconstruction paradigm of Dinomaly (Guo et al. 2025). A frozen DINO encoder extracts multi-level reference features, which are aggregated into the bottleneck representation H . CR-MoE refines H through training-time cross-representation supervision before it is passed to the reconstruction decoder. During hierarchical decoding, CBTM is applied between consecutive stages to model

condition-dependent feature transitions. EUPE and the cross-representation evaluation branches are used only during training, whereas inference retains the DINO encoder, bottleneck, CR-MoE router and experts, decoder, and CBTM components.

Problem Formulation and Motivation

Problem formulation. Given normal-only training data $\mathcal{D}_n = \{x^{(m)}\}_{m=1}^M$, a frozen DINO encoder E_D extracts multi-level reference features (Caron et al. 2021). A bottleneck module aggregates them into $H = B(E_D(x)) = \{h_i\}_{i=1}^N$, $h_i \in \mathbb{R}^C$, and a decoder reconstructs the DINO feature hierarchy. Anomalies are measured in the same space:

$$A(x) = \text{Agg}_{l,i} d(\hat{z}_{l,i}^D, z_{l,i}^D). \quad (1)$$

where d is cosine distance. DINO thus defines both reconstruction targets and the anomaly-localization metric.

Perceptual enrichment without metric distortion. The bottleneck H selectively preserves information aligned with the DINO pre-training objective (Tishby, Pereira, and Bialek 1999; Tishby and Zaslavsky 2015). By the data-processing inequality, a deployable transformation from H cannot recover absent information, but can reorganize accessible cues. We therefore keep DINO as the only inference representation and use frozen EUPE features only to evaluate candidate corrections during training (Zhu et al. 2026). For candidate k at token i ,

$$e_{i,k} = \lambda_D d_{i,k}^D + \lambda_E d_{i,k}^E, \quad (2)$$

where the DINO term constrains fidelity to the anomaly-oriented geometry, while the EUPE term measures compatibility with broader perceptual knowledge. Because supervision acts through candidate evaluation rather than feature fusion, low-capacity experts form a conditional family of compact correction spaces (Jacobs et al. 1991; Shazeer et al. 2017); the subspace and optimization views are in Appendix.

Deployable selection from privileged supervision. EUPE evaluation is privileged information available only during training (Vapnik and Vashist 2009; Lopez-Paz et al. 2016). The expert costs induce a token-wise responsibility distribution

$$q_{i,k} = \frac{\exp(-e_{i,k}/\tau_q)}{\sum_{j=1}^K \exp(-e_{i,j}/\tau_q)}. \quad (3)$$

This is an entropy-regularized expert assignment. A lightweight router predicts it from the DINO bottleneck only, so deployment keeps the part of the selection rule predictable from H . EUPE, projectors, and responsibility generation are removed at inference. The Gibbs view, router optimum, and distillation gap are derived in Appendix.

Conditional recovery under hierarchical ambiguity. Hierarchical reconstruction is a conditional inverse problem: encoder compression is non-invertible, so a deep state X_i does not uniquely determine its shallower counterpart X_{i-1} :

$$H(X_{i-1} | X_i) > 0. \quad (4)$$

A deterministic decoder may average valid recovery modes, while direct shallow skips create a high-bandwidth shortcut

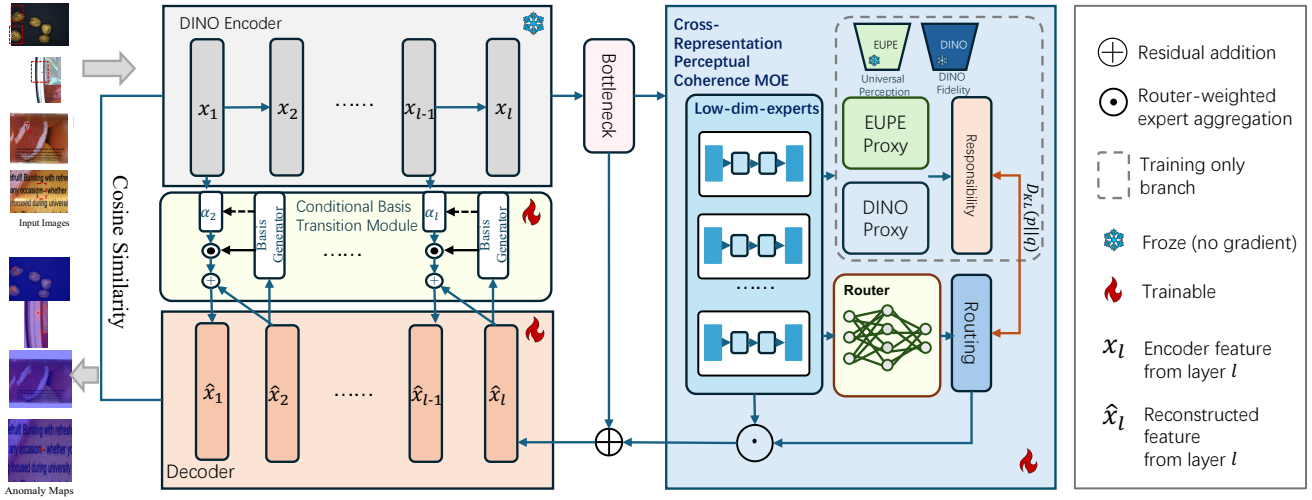


Figure 1: **Overall architecture of the proposed framework.** CR-MoE refines the DINO bottleneck with training-time EUPE supervision, while CBTM models conditional transitions during hierarchical feature reconstruction. At inference, all training-only branches are removed and multi-level DINO reconstruction discrepancies produce the anomaly map.

that weakens the bottleneck constraint (Bishop 1994; Sohn, Lee, and Yan 2015; Geirhos et al. 2020; Yong and Brinrup 2022). We instead represent each layer-wise transition with a state-dependent basis and low-dimensional coefficients:

$$\Delta_{i-1} = B_{i-1}\alpha_{i-1} = \sum_{k=1}^K \alpha_{i-1,k} b_{i-1,k}. \quad (5)$$

The current reconstructed state determines admissible directions, and the adjacent target feature only guides their combination. This constrains transitions without transmitting the target feature to the decoder. Basis regularization, capacity analysis, and architectural details are in Appendix.

These considerations motivate CR-MoE for DINO-only perceptual enrichment and CBTM for constrained layer-wise recovery under conditional ambiguity.

Cross-Representation Perceptual Coherence Mixture-of-Experts

DINO provides a reliable feature space for anomaly localization, but a single representation may miss normal variations in complex industrial scenes. Direct replacement or fusion with heterogeneous encoders can weaken the anomaly-sensitive geometry. As illustrated in Figure 2, CR-MoE learns multiple residual correction directions in the DINO space, evaluates them with DINO and EUPE during training, and distills the selection rule into a DINO-only router for inference.

Residual expert construction. Given $h_i \in \mathbb{R}^C$, K low-capacity residual experts $\{A_k\}_{k=1}^K$ generate candidate states:

$$\begin{aligned} \Delta_{i,k} &= A_k(h_i) = U_k \phi(V_k h_i), \\ h_{i,k} &= h_i + \gamma \Delta_{i,k}. \end{aligned} \quad (6)$$

where $V_k : \mathbb{R}^C \rightarrow \mathbb{R}^r$ and $U_k : \mathbb{R}^r \rightarrow \mathbb{R}^C$ are down- and up-projections with $r \ll C$, and $\phi(\cdot)$ is the activation. The

residual design restricts each expert to a compact correction subspace.

Cross-representation expert evaluation. Each candidate is projected into the DINO and EUPE spaces:

$$\hat{z}_{i,k}^D = P_D(h_{i,k}), \quad \hat{z}_{i,k}^E = P_E(h_{i,k}). \quad (7)$$

where $P_D(\cdot)$ and $P_E(\cdot)$ are projection heads shared across experts, and z_i^D, z_i^E are frozen target features.

Candidate consistency is measured by cosine distance:

$$\begin{aligned} d_{i,k}^D &= 1 - \cos(\hat{z}_{i,k}^D, z_i^D), \\ d_{i,k}^E &= 1 - \cos(\hat{z}_{i,k}^E, z_i^E). \end{aligned} \quad (8)$$

$d_{i,k}^D$ and $d_{i,k}^E$ measure consistency with DINO and EUPE targets, respectively preserving anomaly geometry and adding perceptual evaluation.

The joint expert cost is:

$$e_{i,k} = \lambda_D d_{i,k}^D + \lambda_E d_{i,k}^E. \quad (9)$$

where λ_D and λ_E balance the two evaluation spaces; lower $e_{i,k}$ indicates a more suitable candidate.

Responsibility-guided router distillation. The expert costs define a training-only responsibility distribution:

$$q_{i,k} = \frac{\exp(-e_{i,k}/\tau_q)}{\sum_{j=1}^K \exp(-e_{i,j}/\tau_q)}. \quad (10)$$

where $q_{i,k}$ is expert k 's responsibility for token i , and τ_q controls assignment sharpness. The distribution q_i is training-only.

We distill this cross-representation selection rule into a router that observes only h_i :

$$p_{i,k} = \frac{\exp(G_k(h_i)/\tau_p)}{\sum_{j=1}^K \exp(G_j(h_i)/\tau_p)}. \quad (11)$$

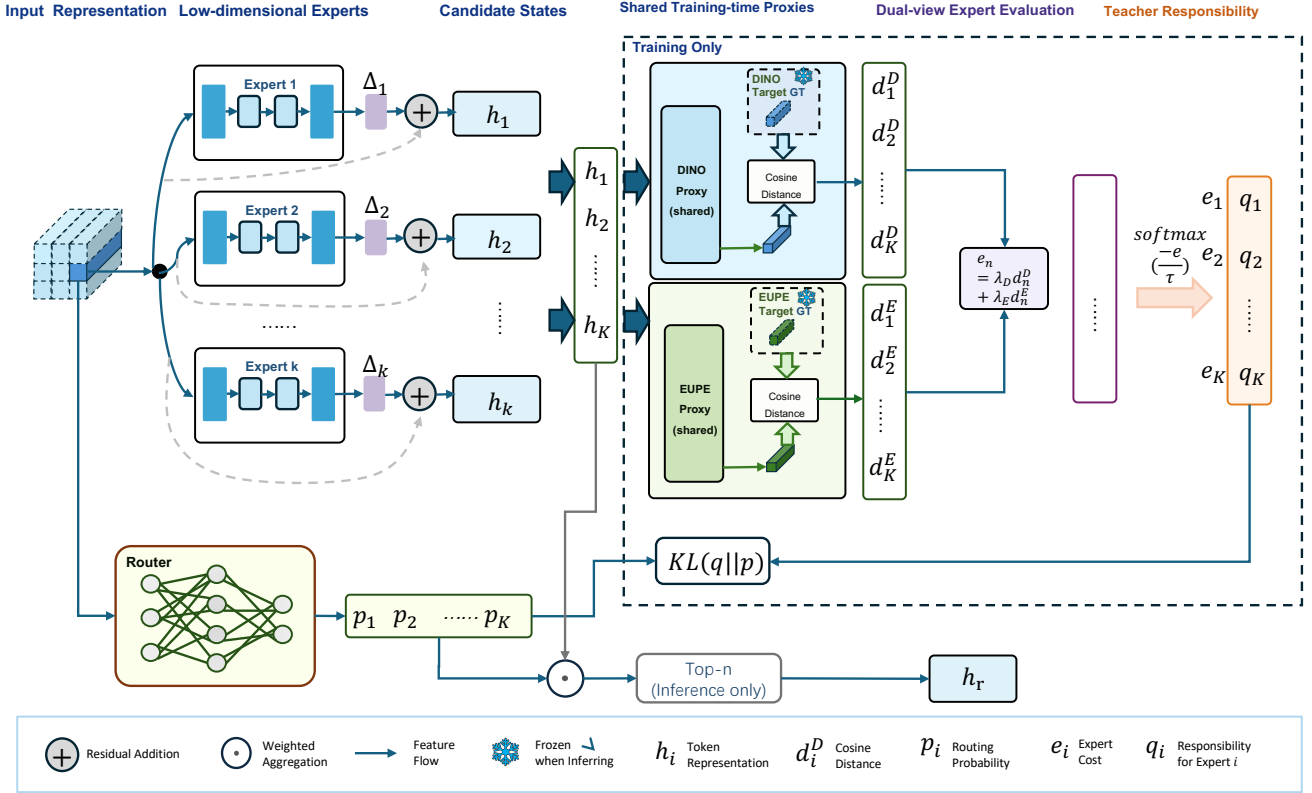


Figure 2: **Workflow of CR-MoE.** Low-dimensional experts generate candidate DINO-space corrections, which are jointly evaluated by DINO and EUPE during training to produce teacher responsibilities. A DINO-only router learns this selection rule and aggregates the Top- n experts at inference.

where $G_k(\cdot)$ is the router logit, $p_{i,k}$ is the routing probability, and τ_p is the routing temperature. Unlike q_i , p_i does not require EUPE at inference.

Conditional expert aggregation. The routed representation is:

$$h_{\text{routed},i} = h_i + \gamma \sum_{k=1}^K p_{i,k} \Delta_{i,k}. \quad (12)$$

Here, $h_{\text{routed},i}$ is passed to the reconstruction decoder. The router combines correction directions rather than heterogeneous features.

For sparse inference, only the Top- n experts under p_i are activated and renormalized; details are in Appendix.

Expert and router optimization. Expert candidates are optimized by:

$$\mathcal{L}_{\text{exp}} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K \text{sg}(q_{i,k}) e_{i,k}. \quad (13)$$

where $\text{sg}(\cdot)$ denotes stop-gradient. The detached responsibility weights each candidate without updating the responsibility branch.

The router is trained by:

$$\mathcal{L}_{\text{route}} = \frac{1}{N} \sum_{i=1}^N D_{\text{KL}}(\text{sg}(q_i) \parallel p_i). \quad (14)$$

This KL loss transfers the training-time responsibility distribution into the DINO-only router.

A weak balancing term prevents routing collapse:

$$\mathcal{L}_{\text{bal}} = \frac{1}{K} \sum_{k=1}^K \left(\bar{p}_k - \frac{1}{K} \right)^2, \quad \bar{p}_k = \frac{1}{N} \sum_{i=1}^N p_{i,k}. \quad (15)$$

Here, $\bar{p}_k$ is the average routing probability of expert k . The

Conditional Basis Transition Module

CR-MoE enriches the bottleneck through cross-representation supervision, but the decoder still performs deterministic stage-wise recovery (Deng and Li 2022; Guo et al. 2025). Since deep features cannot uniquely determine shallow details, similar states may admit different normal recovery paths (Tishby, Pereira, and Bialek 1999; Bishop 1994). CBTM therefore replaces a fixed inter-stage mapping with a sample-conditioned transition, as shown in Figure 3.

Let $x_i \in \mathbb{R}^{N_i \times C_i}$ be the frozen encoder feature at level i , and $\hat{x}_i$ its reconstruction. Levels are indexed from shallow to deep, and decoding follows $\hat{x}_i \rightarrow \hat{x}_{i-1}$.

Conditional basis generation. Given $\hat{x}_i$, the Basis Generator produces K transition bases:

$$B_{i-1} = G_{i-1}^B(\hat{x}_i) = [b_{i-1,1}, \dots, b_{i-1,K}]. \quad (16)$$

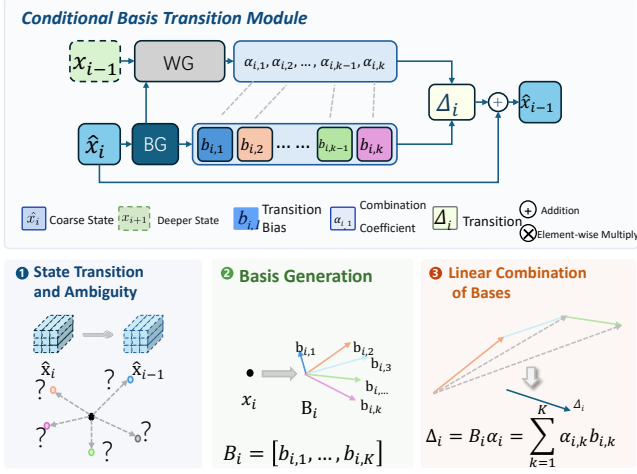


Figure 3: **Workflow of CBTM.** The current reconstructed state generates a compact set of transition bases, while the corresponding encoder feature predicts their combination coefficients. The resulting conditional transition is added to the base decoder output for hierarchical feature recovery.

Here, $G_{i-1}^B(\cdot)$ is a two-layer MLP. $B_{i-1} \in \mathbb{R}^{N_{i-1} \times K \times C_{i-1}}$ defines a local transition space, and each $b_{i-1,k}$ is one candidate direction (Mairal et al. 2009; Jia et al. 2016).

Transition coefficient prediction. The encoder feature x_{i-1} predicts token-wise coefficients:

$$\alpha_{i-1} = G_{i-1}^C(x_{i-1}) = [\alpha_{i-1,1}, \dots, \alpha_{i-1,K}]. \quad (17)$$

Here, $G_{i-1}^C(\cdot)$ is another two-layer MLP, and $\alpha_{i-1} \in \mathbb{R}^{N_{i-1} \times K}$ specifies token-wise basis weights. The encoder feature acts as a selection signal rather than a direct reconstruction input (Shazeer et al. 2017; Han et al. 2022).

The conditional transition is computed as

$$\Delta_{i-1} = B_{i-1} \alpha_{i-1} = \sum_{k=1}^K \alpha_{i-1,k} b_{i-1,k}. \quad (18)$$

Here, $\Delta_{i-1} \in \mathbb{R}^{N_{i-1} \times C_{i-1}}$ is the token-wise transition. The deep state gives available directions, and the target-level feature selects their combination.

Hierarchical state recovery. The decoder first produces a base reconstruction:

$$\bar{x}_{i-1} = D_{i-1}(\hat{x}_i), \quad (19)$$

where $D_{i-1}(\cdot)$ is the decoder block and $\bar{x}_{i-1}$ is its output.

The final reconstructed state is

$$\hat{x}_{i-1} = \bar{x}_{i-1} + \Delta_{i-1}. \quad (20)$$

The resulting feature is passed to the next shallower stage, yielding $\hat{x}_L \rightarrow \hat{x}_{L-1} \rightarrow \dots \rightarrow \hat{x}_1$. The base decoder gives the main recovery, while CBTM adds a constrained sample-dependent transition.

Basis orthogonality regularization. The bases and coefficients are learned without direct supervision. To reduce redundant directions, we use an orthogonality regularizer (Bansal, Chen, and Wang 2018; Cogswell et al. 2016).

For token n at level $i-1$, let $B_{i-1,n} \in \mathbb{R}^{K \times C_{i-1}}$ denote its basis matrix. Each basis vector is normalized as

$$\tilde{b}_{i-1,n,k} = \frac{b_{i-1,n,k}}{\|b_{i-1,n,k}\|_2 + \epsilon}, \quad (21)$$

where ϵ ensures numerical stability, forming $\tilde{B}_{i-1,n} \in \mathbb{R}^{K \times C_{i-1}}$.

The orthogonality loss is defined as

$$\mathcal{L}_{\text{orth}} = \frac{1}{L-1} \sum_{i=2}^L \frac{1}{N_{i-1}} \sum_{n=1}^{N_{i-1}} \left\| \tilde{B}_{i-1,n} \tilde{B}_{i-1,n}^\top - I_K \right\|_F^2. \quad (22)$$

Here, I_K is the identity matrix and $\|\cdot\|_F$ denotes the Frobenius norm. The off-diagonal terms penalize correlated bases.

The CBTM-specific regularization is

$$\mathcal{L}_{\text{CBTM}} = \lambda_{\text{orth}} \mathcal{L}_{\text{orth}}, \quad (23)$$

where λ_{orth} controls regularization strength. The Basis Generator and Coefficient Predictor are jointly optimized with the reconstruction objective.

Optimization and Anomaly Inference

Following Dinomaly (Guo et al. 2025), we supervise reconstruction over the selected DINO feature levels using the cosine-distance objective

$$\mathcal{L}_{\text{rec}} = \frac{1}{L} \sum_{l=1}^L \frac{1}{N_l} \sum_{i=1}^{N_l} [1 - \cos(\hat{x}_{l,i}, x_{l,i})], \quad (24)$$

where x_l and $\hat{x}_l$ denote the reference and reconstructed features at level l , respectively. The complete objective combines the reconstruction loss with the module-specific objectives defined before.

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{CR}} \mathcal{L}_{\text{CR-MoE}} + \lambda_{\text{CBT}} \mathcal{L}_{\text{CBTM}}. \quad (25)$$

The DINO and EUPE encoders remain frozen. The bottleneck, decoder, residual experts, router, basis generators, coefficient predictors, and auxiliary projection heads are optimized jointly. EUPE and all cross-representation evaluation components are discarded after training.

At inference, anomaly evidence is computed entirely in the DINO feature space. Following the loose reconstruction strategy of Dinomaly, the layer-wise cosine-distance maps are spatially aligned, upsampled to the input resolution, and aggregated:

$$\mathcal{A}(x) = \text{Agg}_{l=1}^L \text{Up}(1 - \cos(\hat{x}_l, x_l)). \quad (26)$$

Experiments

Experimental Setup

We evaluate one normal-only model per dataset on MVTEC AD (Bergmann et al. 2019), VisA (Zou et al. 2022), RealIAD (Wang et al. 2024), BTAD (Mishra et al. 2021), and MVTEC AD 2 (Heckler-Kram et al. 2026); AD2 uses the public fold and 512×512 tiles with 10% overlap. We report I-F1, P-AUPRO@0.05, and P-AP. P-AUPRO@0.05 weights defect

Plain	U-Net	Skip	Fixed	CBTM	APG $\uparrow$
✓					0.074
	✓				0.068
		✓			0.036
			✓		0.081
				✓	0.094

Table 1: Anomaly-preservation gap under alternative recovery paths.

regions equally within the low-false-positive range (Heckler-Kram et al. 2026), and I-F1 is the maximum F1 over image-level operating points. RD4AD, Dinomaly, INP-Former, and RAD (Deng and Li 2022; Guo et al. 2025; Luo et al. 2025; Zhang et al. 2026) are reproduced with identical splits, pre-processing, training budget, post-processing, scoring, and metrics.

The frozen backbones are DINOv2 ViT-B/14 Reg4 and EUPE-ViT-B/16. Inputs are resized to 448^2 and center-cropped to 392^2 , giving 28^2 DINO tokens. DINO targets average layers [2, 3, 4, 5] and [6, 7, 8, 9]; EUPE uses [6, 7, 8, 9]. CR-MoE uses four rank-64 experts, Top-2 routing, $(\tau_p, \tau_q) = (0.5, 0.003)$, DINO/EUPE weights 1.0/0.15, and residual scale γ initialized to 0.1. CBTM uses eight rank-8 bases, two transitions, and coefficient bound 2.0; basis and coefficient predictors read decoder and encoder features, respectively. We set $(\lambda_{CR}, \lambda_{CBT}, \lambda_{orth}) = (0.1, 1, 10^{-3})$. StableAdamW trains for 10 epochs with batch size 24, cosine LR $2 \times 10^{-3} \rightarrow 2 \times 10^{-4}$, 100-step warmup, weight decay 10^{-4} , and gradient clipping 0.1. Inference averages two residual maps, resizes them to 256^2 , applies 5×5 Gaussian smoothing ($\sigma = 4$), and averages the top 1% pixels for image scoring; EUPE is removed. Experiments run on one NVIDIA GeForce RTX 5090 (32 GB), 25 Intel Xeon Platinum 8470Q vCPUs, and 90 GB RAM with Ubuntu 22.04, Python 3.12, PyTorch 2.5.1, and CUDA 12.4.

Comparison with Representative Methods

Table 2 compares image-level detection and low-false-positive pixel localization across five datasets.

Our method ranks first in 11 of 15 comparisons. Against the strongest baseline, P-AUPRO@0.05 improves by 5.8%, 3.5%, 1.0%, 1.8%, and 3.9% on MVTec AD, VisA, Real-IAD, BTAD, and AD2. It also leads P-AP and I-F1 on VisA, Real-IAD, and AD2, while remaining comparable elsewhere. The consistent P-AUPRO@0.05 gains show that the main advantage lies in localization under a low false-positive budget rather than saturated image-level discrimination. For the remaining close comparisons, MoReAD reaches 75.5 P-AP on MVTec AD versus 75.6 for RAD, and 71.6 on BTAD versus 71.8 for INP-Former.

Figure 4 shows that the baseline responds to normal high-contrast package structures, whereas MoReAD removes most scattered responses and retains a compact activation at the annotated printing error.

EUPE is absent from inference. Compared with DINO, MoReAD adds 1.0% parameters and 4.9% memory; dense

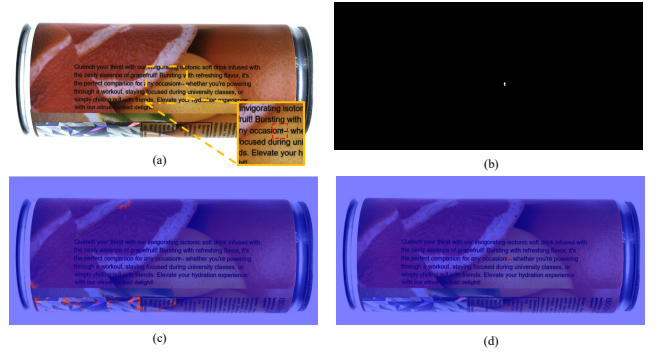


Figure 4: A subtle printing defect in a text-rich region: (a) input and magnified defect, (b) ground truth, (c) baseline map, and (d) MoReAD map.

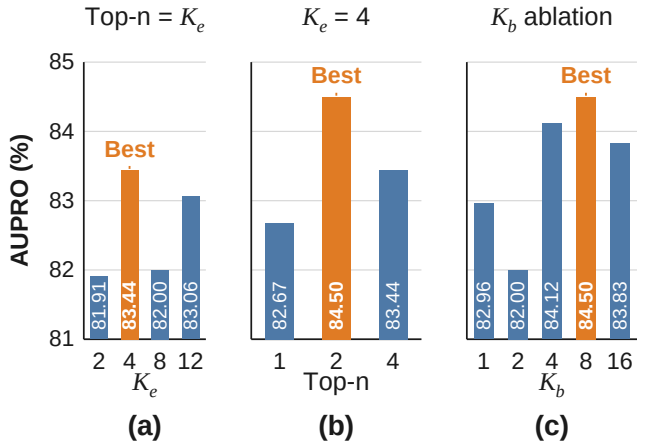


Figure 5: Sensitivity on the VisA-Real-IAD mean. All datasets use the same configuration: $K_e = 4$, Top- $n = 2$, and $K_b = 8$.

token-wise routing and transitions increase GFLOPs and latency by 47.6% and 55.0%, reducing FPS by 35.5% (Table 3).

Ablation and Mechanism Analysis

Ablations average dataset-level scores on VisA and Real-IAD. Tables 4 and 5 test the two design hypotheses. We use $\text{Geo} = \mathbb{E}_{i,j} |\cos(z_i, z_j) - \cos(h_i, h_j)|$ to measure DINO geometry drift between reference and transformed normal tokens, and $\text{APG} = \mathbb{E}[d | \text{anomaly}] - \mathbb{E}[d | \text{normal}]$ to measure reconstruction-distance separation. Lower Geo and higher APG are preferable.

Cross-representation transfer. Direct DINO-EUPE fusion reduces P-AUPRO@0.05/P-AP from 80.37/60.71 to 69.20/49.60, confirming distortion of the fixed scoring space (Table 4). DINO-space experts reach 81.70/62.15, and EUPE evaluation reaches 83.05/63.60. The DINO-only router retains 82.95/63.48 after EUPE removal, supporting training-only cross-representation evaluation without changing the inference metric. Meanwhile, I-F1 stays within 97.18–97.38,

Method	MVTec AD	VisA	Real-IAD	BTAD	MVTec AD2
RD4AD (2022)	95.2/75.2/48.6	89.6/78.2/38.0	73.9/77.3/25.0	93.8/55.2/57.1	50.9/26.6/17.4
Dinomaly (2025)	99.0/79.7/69.3	96.2/80.9/53.2	80.2/82.1/42.8	95.6 /57.4/70.1	84.3/59.9/37.0
INP-Former (2025)	99.2 /79.5/71.0	96.6/82.3/51.2	81.5/82.7/47.5	95.2/56.3/ 71.8	80.3/65.2/39.5
RAD (2026)	99.0/79.9/ 75.6	95.2/—/55.3	83.0/84.1/53.7	95.5/61.2/71.5	84.3/66.7/34.3
Ours	99.1/ 84.5 /75.5	96.8 / 85.2 / 56.4	83.1 / 84.9 / 53.9	95.3/ 62.3 /71.6	85.0 / 69.3 / 43.9

Table 2: Multi-class results (%). Each cell reports I-F1/P-AUPRO@0.05/P-AP.

Method	Training branches	Inference branches	Train. params	Infer. params	GFLOPs	Memory (MiB)	FPS	Latency (ms)
DINO baseline	DINO	DINO	61.43M	148.01M	210.62	667.1	159.59	6.27
EUPE inference	DINO+EUPE	DINO+EUPE	61.43M	233.68M	346.03	993.6	90.85	11.02
CR-MoE	DINO+EUPE eval.	DINO+CR-MoE	63.00M	148.41M	309.00	677.0	135.30	7.39
CBTM	DINO+EUPE eval.+fixed CBTM	DINO+CR-MoE+fixed CBTM	63.92M	149.32M	310.46	676.1	117.88	8.48
MoReAD	DINO+EUPE+CR-MoE+CBTM	DINO+CR-MoE+CBTM	64.14M	149.54M	310.81	699.8	102.90	9.72

Table 3: Model complexity and inference efficiency.

Direct fusion	MoE	EUPE evaluator	DINO router	I-F1↑	P-AUPRO@0.05↑	P-AP↑
				97.25	80.37	60.71
				97.18	81.70	62.15
	✓			(−0.07)	(+1.33)	(+1.44)
				97.38	83.05	63.60
	✓	✓		(+0.20)	(+1.35)	(+1.45)
				97.31	82.95	63.48
	✓	✓	✓	(−0.07)	(−0.10)	(−0.12)
✓				93.10	69.20	49.60

Table 4: Cross-representation transfer and router-distillation ablation on VisA and Real-IAD (%). Parentheses denote changes from the preceding nested configuration.

CR-MoE	U-Net	Direct skip	Fixed basis	CBTM (full)	I-F1↑	P-AUPRO@0.05↑	P-AP↑
					97.25	80.37	60.71
	✓				97.10	80.25	60.60
✓		✓			95.90	76.80	55.40
✓			✓		97.22	83.25	64.10
✓				✓	97.45	84.85	65.70
					(+0.23)	(+1.60)	(+1.60)

Table 5: Hierarchical feature-recovery ablation on VisA and Real-IAD (%). The first row uses the plain decoder; parentheses report changes from the fixed-basis configuration.

showing that the improvement is concentrated in fine localization rather than image-level decisions.

Constrained hierarchical recovery. A capacity-matched U-Net (Ronneberger, Fischer, and Brox 2015) matches the plain decoder, whereas direct skips impair all metrics (Table 5). Bounded coefficients recover localization, and conditional rather than fixed bases increase P-AUPRO@0.05 from 83.25 to 84.85 and P-AP from 64.10 to 65.70. This supports compact, conditional hierarchical transitions over full feature transfer. The U-Net comparison separates transition model-

Direct fusion	MoE	EUPE evaluator	Geo↓
			0.269
	✓		0.145
		✓	0.110
✓			0.327

Table 6: DINO geometry drift under cross-representation transfer.

ing from decoder capacity, while the direct-skip degradation shows that full feature transfer is not a substitute for the proposed transition structure.

Mechanism diagnostics. Candidate evaluation lowers Geo from 0.269 to 0.110, whereas direct fusion raises it to 0.327 (Table 6). Fixed and conditional bases raise APG from the direct-skip value of 0.036 to 0.081 and 0.094 (Table 1). Figure 5 favors moderate expert and basis counts with Top-2 routing. The agreement between task metrics and both diagnostics links the gains to geometry-preserving transfer and constrained hierarchical recovery.

Conclusion

We present a feature-reconstruction framework that addresses representation incompleteness and hierarchical recovery ambiguity in industrial anomaly detection. CR-MoE transfers universal perceptual knowledge through training-time cross-representation evaluation while retaining a DINO-only inference space. CBTM models layer-wise normal recovery using constrained conditional transitions. They improve normal reconstruction without unrestricted feature shortcuts or additional inference encoders. Experiments demonstrate consistent gains in strict pixel-level localization.

Future work may explore more efficient and input-dependent expert and basis allocation.

References

- Alemi, A. A.; Fischer, I.; Dillon, J. V.; and Murphy, K. 2017. Deep Variational Information Bottleneck. In *International Conference on Learning Representations*.
- Amir, S.; Gandelsman, Y.; Bagon, S.; and Dekel, T. 2021. Deep ViT Features as Dense Visual Descriptors. *arXiv preprint arXiv:2112.05814*.
- Bansal, N.; Chen, X.; and Wang, Z. 2018. Can We Gain More from Orthogonality Regularizations in Training Deep Networks? In *Advances in Neural Information Processing Systems*, volume 31, 4266–4276.
- Beizae, F.; Lodygensky, G. A.; Desrosiers, C.; et al. 2025. Correcting Deviations from Normality: A Reformulated Diffusion Model for Multi-Class Unsupervised Anomaly Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 19088–19097.
- Bergmann, P.; Fauser, M.; Sattlegger, D.; and Steger, C. 2019. MVTec AD—A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 9592–9600.
- Bishop, C. M. 1994. Mixture Density Networks. Technical Report NCRG/94/004, Neural Computing Research Group, Aston University.
- Caron, M.; Touvron, H.; Misra, I.; Jégou, H.; Mairal, J.; Bojanowski, P.; and Joulin, A. 2021. Emerging Properties in Self-Supervised Vision Transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 9650–9660.
- Cogswell, M.; Ahmed, F.; Girshick, R.; Zitnick, C. L.; and Batra, D. 2016. Reducing Overfitting in Deep Networks by Decorrelating Representations. In *International Conference on Learning Representations*.
- Deng, H.; and Li, X. 2022. Anomaly Detection via Reverse Distillation from One-Class Embedding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 9737–9746.
- Geirhos, R.; Jacobsen, J.-H.; Michaelis, C.; Zemel, R.; Brendel, W.; Bethge, M.; and Wichmann, F. A. 2020. Shortcut Learning in Deep Neural Networks. *Nature Machine Intelligence*, 2(11): 665–673.
- Guo, J.; Lu, S.; Jia, L.; Zhang, W.; and Li, H. 2023. Re-contrast: Domain-specific anomaly detection via contrastive reconstruction. *Advances in Neural Information Processing Systems*, 36: 10721–10740.
- Guo, J.; Lu, S.; Zhang, W.; Chen, F.; Li, H.; and Liao, H. 2025. Dinomaly: The Less Is More Philosophy in Multi-Class Unsupervised Anomaly Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 20405–20415.
- Han, Y.; Huang, G.; Song, S.; Yang, L.; Wang, H.; and Wang, Y. 2022. Dynamic Neural Networks: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(11): 7436–7456.
- Heckler-Kram, L.; Neudeck, J.-H.; Scheler, U.; König, R.; and Steger, C. 2026. The mvtec ad 2 dataset: Advanced scenarios for unsupervised anomaly detection. *International Journal of Computer Vision*, 134(4): 175. Article 175.
- Jacobs, R. A.; Jordan, M. I.; Nowlan, S. J.; and Hinton, G. E. 1991. Adaptive Mixtures of Local Experts. *Neural Computation*, 3(1): 79–87.
- Jia, X.; De Brabandere, B.; Tuytelaars, T.; and Van Gool, L. 2016. Dynamic Filter Networks. In *Advances in Neural Information Processing Systems*, volume 29.
- Liu, J.; Xie, G.; Wang, J.; Li, S.; Wang, C.; Zheng, F.; and Jin, Y. 2024. Deep industrial image anomaly detection: A survey. *Machine Intelligence Research*, 21(1): 104–135.
- Liu, X.; Wang, J.; Leng, B.; and Zhang, S. 2025. Unlocking the Potential of Reverse Distillation for Anomaly Detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, 5640–5648.
- Lopez-Paz, D.; Bottou, L.; Schölkopf, B.; and Vapnik, V. 2016. Unifying Distillation and Privileged Information. In *International Conference on Learning Representations*.
- Luo, W.; Cao, Y.; Yao, H.; Zhang, X.; Lou, J.; Cheng, Y.; Shen, W.; and Yu, W. 2025. Exploring intrinsic normal prototypes within a single image for universal anomaly detection. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, 9974–9983.
- Mairal, J.; Bach, F.; Ponce, J.; and Sapiro, G. 2009. On-line Dictionary Learning for Sparse Coding. In *Proceedings of the 26th Annual International Conference on Machine Learning*, 689–696.
- Mishra, P.; Verk, R.; Fornasier, D.; Piciarelli, C.; and Foresti, G. L. 2021. VT-ADL: A Vision Transformer Network for Image Anomaly Detection and Localization. In *2021 IEEE 30th International Symposium on Industrial Electronics (ISIE)*, 1–6.
- Oquab, M.; Darcet, T.; Moutakanni, T.; Vo, H. V.; Szafraniec, M.; Khalidov, V.; Fernandez, P.; Haziza, D.; Massa, F.; El-Nouby, A.; Assran, M.; Ballas, N.; Galuba, W.; Howes, R.; Huang, P.-Y.; Li, S.-W.; Misra, I.; Rabbat, M.; Sharma, V.; Synnaeve, G.; Xu, H.; Jégou, H.; Mairal, J.; Labatut, P.; Joulin, A.; and Bojanowski, P. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*.
- Ronneberger, O.; Fischer, P.; and Brox, T. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 234–241. Springer.
- Salehi, M.; Sadjadi, N.; Baselizadeh, S.; et al. 2021. Multiresolution knowledge distillation for anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 14902–14912.
- Shazeer, N.; Mirhoseini, A.; Maziarz, K.; Davis, A.; Le, Q. V.; Hinton, G. E.; and Dean, J. 2017. Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer. In *International Conference on Learning Representations*.

Sohn, K.; Lee, H.; and Yan, X. 2015. Learning Structured Output Representation Using Deep Conditional Generative Models. In *Advances in Neural Information Processing Systems*, volume 28, 3483–3491.

Tao, X.; Gong, X.; Zhang, X.; Yan, S.; and Adak, C. 2022. Deep learning for unsupervised anomaly localization in industrial images: A survey. *IEEE Transactions on Instrumentation and Measurement*, 71: 1–21.

Tian, Y.; Sun, C.; Poole, B.; Krishnan, D.; Schmid, C.; and Isola, P. 2020. What Makes for Good Views for Contrastive Learning? In *Advances in Neural Information Processing Systems*, volume 33, 6827–6839.

Tien, T.-D.; Nguyen, A.-T.; Tran, N.-H.; et al. 2023. Revisiting reverse distillation for anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 24511–24520.

Tishby, N.; Pereira, F. C.; and Bialek, W. 1999. The Information Bottleneck Method. In *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*, 368–377.

Tishby, N.; and Zaslavsky, N. 2015. Deep Learning and the Information Bottleneck Principle. In *2015 IEEE Information Theory Workshop*, 1–5. IEEE.

Vapnik, V.; and Vashist, A. 2009. A New Learning Paradigm: Learning Using Privileged Information. *Neural Networks*, 22(5–6): 544–557.

Wang, C.; Zhu, W.; Gao, B.-B.; Gan, Z.; Zhang, J.; Gu, Z.; Qian, S.; Chen, M.; and Ma, L. 2024. Real-iad: A real-world multi-view dataset for benchmarking versatile industrial anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 22883–22892.

Xie, G.; Wang, J.; Liu, J.; Lyu, J.; Liu, Y.; Wang, C.; Zheng, F.; and Jin, Y. 2024. IM-IAD: Industrial Image Anomaly Detection Benchmark in Manufacturing. *arXiv preprint arXiv:2301.13359*. Version 5, last revised January 2024.

Yong, B. X.; and Brintrup, A. 2022. Do Autoencoders Need a Bottleneck for Anomaly Detection? *IEEE Access*, 10: 78455–78471.

Zavrtanik, V.; Kristan, M.; and Skocaj, D. 2021. DR-AEM – A discriminatively trained reconstruction embedding for surface anomaly detection. *arXiv preprint arXiv:2108.07610*.

Zhang, X.; Li, G.; Henderson, P.; Aragon-Camarasa, G.; and Long, Z. 2026. Is Task-Specific Training Necessary for Anomaly Detection? In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306.

Zhu, C.; Suri, S.; Jose, C.; Oquab, M.; Szafraniec, M.; Wen, W.; Xiong, Y.; Labatut, P.; Bojanowski, P.; Krishnamoorthi, R.; and Chandra, V. 2026. Efficient Universal Perception Encoder. *arXiv preprint arXiv:2603.22387*.

Zou, Y.; Jeong, J.; Pemula, L.; Zhang, D.; and Dabeer, O. 2022. SPot-the-Difference Self-Supervised Pre-training for Anomaly Detection and Segmentation. In *Computer Vision—ECCV 2022*, volume 13690 of *Lecture Notes in Computer Science*, 392–408. Springer, Springer Nature Switzerland.